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Estimation of pyrolysis liquid product yield and its hydrogen content for biomass resources by combined evaluation of pyrolysis conditions with proximate-ultimate analysis data: A machine learning application

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ABSTRACT

This study, it was aimed to predict the pyrolysis liquid product yield (Liquid P. Yield (%wt.)) and its hydrogen content (H-oil (%wt.)) based on the biomass composition and pyrolysis conditions. For this purpose, firstly, serial previous studies on biomass pyrolysis in literature were collected in a data set, and the data set used in the study was presented in the Supplementary information document. Then machine learning application was made. Within the scope of the study, Multiple Linear Regression Theory (MLR), Decision Three Theory (DT), Gaussian Model Theory (GM), and Random Forest Theory (RF) were applied to the data set obtained from the literature, and the success rates were compared. As a result, it has been determined that the RF model provides more successful results than other models in predicting the pyrolysis liquid product yield and hydrogen content. In addition, the importance feature information was obtained by using the average pollution reduction method in the whole models of RF. A detailed partial dependency analysis was performed to investigate in-depth the relationship between biomass composition and pyrolysis conditions and target variables. Thus, the amount of liquid product and the hydrogen content (quality) of the liquid product was optimized depending on the biomass composition and pyrolysis conditions. In addition, it was determined that the biomass composition was more effective on the liquid product yield and H-content than the pyrolysis conditions. It is thought that the results and comments of this study will guide the researchers in the process of pyrolysis of different biomass wastes that emerge seasonally and contribute to the more effective evaluation of biomass resources. In addition, The RF model can be used as a good reference for the modeling studies on biomass pyrolysis liquid product distribution and its quality.

1. Introduction

1.1. Biomass resources and clean energy

Biomass covers all organic matter (aquatic or terrestrial plants, animal, urban or food industry wastes, and agricultural and forest by-products.), that can be regenerated in less than 100 years, although biomass was originally thought of only as plant organisms, that stored solar energy with the help of photosynthesis [1–3]. Biomass energy resources are compared to fossil energy sources, due to containing much lower amounts of nitrogen and sulfur and the lower combustion temperature compared to fossil fuels, NO_x emissions are very low [4,5] In addition, the biomass energy sources are carbon-neutral energy sources. There are no net CO_2 emissions to the atmosphere during the combustion process. Therefore, they do not cause global warming and acid rain. Thus, the energy obtained using biomass resources (bio-gas, bio-oil, bio-ethanol) is called green energy and clean energy. Biomass energy has strategic importance as it is both renewable and environmentally friendly [6].

1.2. Biomass reserves in Turkey and bio-energy potential of Turkey

It produces a wide variety of agricultural and forestry products in Turkey. A large amount of waste is generated as a result of agricultural and forestry activities. According to BEPA, Turkey's energy potential atlas (2021)[7], Turkey has approximately 100 million tons of biomass waste reserves annually (Table 1). To create more income from these agricultural and forestry products, protect the environment and nature, and promote socio-economic development, these wastes must be collected and evaluated as bioenergy/bio-refinery raw material.

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Table 1
Waste biomass reserves in Turkey and bio-energy potential of Turkey (BEPA, Turkey's energy potential atlas, 2021).

Agricultural Production Amount (Ton/Year)	171.399.002
Agricultural Waste Amount (Ton/Year)	62.206.754
Theoretical Energy Equivalent of Agricultural Wastes (TEP/Year)	25.384.268
Economic Energy Equivalent of Agricultural Wastes (TEP/Year)	1.462.159
Amount of Municipal Waste (Ton/Year)	32.170.975
Theoretical Energy Equivalents of Municipal Wastes (TEP/Year)	3.373.011
Economic Energy Equivalents of Municipal Wastes (TEP/Year)	485.858
Forest Asset Residues (Ster/Year)	3.914.904
Energy Equivalent of Forest Residues (TEP/Year)	859.899

However, these wastes have different structural-chemical compositions and calorific values. To evaluate these seasonal wastes more effectively, it is necessary to create suitable combined mixtures from waste biomass sources and to feed them to the thermal conversion processes.

1.3. The necessity of conversion processes and the advantage of the pyrolysis process

Biomass resources are generally inhomogeneous, with high moisture and oxygen content, low density, and low calorific value [8,9]. These properties negatively affect the fuel qualities and the cycle efficiency of the combustion processes are quite low. The existing negative features of biomass resources have to be improved by conversion processes (physical processes (drying, grinding, pelleting, compounding, filtration, extraction, etc.), thermochemical (pyrolysis, combustion, gasification, or liquefaction), or biochemical (fermentation, bio-photolysis, or anaerobic digestion processes). In conversion processes applied to biomass, the main aim is to obtain alternatives to fossil fuels, stable, easily storable, and portable solid, liquid, or gaseous fuels, or to produce valuable chemicals for the chemical industry. Pyrolysis is a thermal conversion process that uses the principle of direct heat (thermal) degradation of an organic matrix such as biomass waste in the absence of oxygen to produce liquid (oil/tar), gaseous, and solid (char) products [10].

Many factors affect product quality, composition, and yield in the pyrolysis process. These parameters are the pyrolysis conditions (heating rate (HR), pyrolysis final temperature (HTT), the flow ratio of the inert atmosphere (FR), particle size (PS), residence time, the structure of the raw material, etc.) and the composition of the biomass source (ultimate, proximate and chemical composition of biomass resources). These parameters are optimized to achieve the desired amount of target product.

1.4. The motivation of this study and the gap in the literature

Proximate composition (volatile matter, ash, moisture, and fixed carbon) and ultimate composition (C, H, S, N, and O) of biomass resources can be determined using cheap and easy methods that can be done easily and in a short time under laboratory conditions. In addition, these analyses do not require a special device, equipment, or chemical. Thus, it was thought that by utilizing the structural composition of the biomass resources, and the pyrolysis conditions, estimating the amount of pyrolysis liquid product and H-content will be an advantage. For this purpose, it was thought that regression methods (linear and non-linear) could be used to detect correlations between variables. In this way, it will be possible to evaluate seasonal waste biomass sources with different compositions together. Combined raw materials suitable for thermal conversion processes will be prepared. Considering Turkey's seasonally changing agricultural and forestry waste species diversity and reserves, it is thought that it will pave the way for efficient and sustainable utilization of wastes.

The most popular and straightforward method for detecting a correlation between variables is linear regression [11,12]. However, linear regression performance is unacceptably low if there is a non-linear correlation between variables (such as biomass composition and fluid yield/quality of pyrolysis condition and fluid yield/quality). Therefore, in recent years, researchers have turned to artificial intelligence and machine learning applications. In this way, many new methods; least squares support vector machine, artificial neural network (ANN), Levenberg-Marquardt ANN method, Decision tree (DT), Gaussian method (GM), and Random forest (RF) are included in traditional engineering research. And thus, acceptable predictive results were obtained [13–17]. Some of the models (DT, GM, and RL methods) expressed in this study were evaluated in the prediction of biomass pyrolysis liquid product yield and quality. It has been tried to reveal that the target variables can be estimated with a certain percentage accuracy by evaluating the data set data. Model successes were compared and the most successful model was tried to be determined.

2. Methods

2.1. Pretreatment and data sets

Two data sets were used in the study. These are; the pyrolysis liquid product (%wt.) data set and its hydrogen content (%wt) data set. The two data sets used in the study were also collected from studies in the literature. The liquid product yield (%wt.) dataset consists of 238 samples, and the hydrogen content of the liquid product dataset consists of 60 samples. Details of the datasets are presented in the "Supplementary document" in tabular form. In this study, the hydrogen content was used as the quality parameter of the pyrolysis liquid product. In general the main issue for the pyrolysis liquid product is the oxygen content as it reduces the energetic efficiency. However, the oxygen content is not directly determined in the ultimate analysis. It is calculated indirectly with the help of the equation. For this reason, it is predicted that the percent error and deviation values are high according to the hydrogen content. Considering this situation, it was considered more meaningful to use the hydrogen contents of the liquid product as a comparison parameter.

- To better demonstrate and interpret the impact of biomass composition and pyrolysis conditions on pyrolysis liquid product (LP) yield and quality (H-content), datasets were generated from different biomass species (herbaceous plant, lignocellulosic biomass, and algae).
- To increase the accuracy/precision of the data analysis, only the studies carried out in the fixed bed pyrolysis experimental setup were taken into account.
- Studies in the last twenty years (i.e. studies done after 2000) are included in the data set in tables.
- Each piece of data contained valid values for all the variables.
- In this study, the composition of biomass resources and pyrolysis condition data were used as the method input, while the liquid product yield and its hydrogen content were acquired as the output.
- The yield of the liquid product (%wt.) and the hydrogen content of the liquid product (H-oil (%wt.)) are presented on a dry basis.
- The composition properties of biomass resources are included the proximate (ie. ash, fixed carbon, volatile matter content), and ultimate analysis data (ie. H, C, N, and O content).
- The data presented in the dataset shows the effects of final pyrolysis temperature (HTT), heating rate (HT) and particle size (PS), an inert atmosphere flow rate (FR) on pyrolysis liquid product yield (%wt.), and H-content of liquid product gathered to reveal.
- All pyrolysis experiments presented in the dataset were performed in an inert environment. Studies that included nitrogen flow rate (FR) information instead of residence time were collected to collect more dataset samples. But, it is possible to evaluate the residence time from the final temperature and heating rate data.

To represent datasets in this study, a box plot was created (including minimum and maximum values, mean, quartile, median, and bar distribution). To determine the linear relationship between any two factors in the datasets, the Pearson correlation coefficient (PCC) was calculated. Eq. (1) presented below was used in the calculation.

$$r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} \tag{1}$$

$Cov(X,Y)$ denotes the covariance between Y and X. The standard deviations of Y and X are represented by σ_Y and σ_X respectively. The second stage was to compute a statistical value using Eq. (2), which proved to be subject to a two-tailed t-distribution with (N-2) degrees of freedom.

$$t = \frac{r}{\sqrt{\frac{1-r^2}{N-2}}} \tag{2}$$

where r is the PCC and N denotes the number of related samples.

Then, the p-value of the significance level indicator was revealed. Because the liquid product yield (percent wt.) dataset and the H-content of the liquid dataset (percent wt.) had different sample counts, PCCs between inputs were determined from the comprehensive dataset. Outputs were produced from subsets (namely, the liquid product yield (percent wt.) dataset and the liquid product hydrogen content dataset). If there is a large PCC between the two variables, either will be removed from size reduction, particularly for linear regression.

2.2. Regression methods and evaluation indicators

2.2.1. Multiple linear regression

Multiple linear regression (MLR) is a fundamental regression technique for identifying linear relationships between several variables. If the relevant performance achieves an acceptable level, it has high interpretability. To obtain convincing accuracy, the approach described in Eqs. (3) and (4) is used in the calculation of error matrix and coefficient vector instead of the guard descent method.

$$X_b = \begin{pmatrix} 1 & X_1^{(1)} & \dots & X_n^{(1)} \\ 1 & X_1^{(2)} & \dots & X_2^{(2)} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ 1 & X_1^{(n)} & \dots & X_m^{(n)} \end{pmatrix} \tag{3}$$

$$\theta = (\theta_0 \quad \theta_1 \dots \quad \theta_n)^T = \left(X_b^T * X_b\right)^{-1} X_b^T \quad y \tag{4}$$

X_b being the feature matrix; $X_i^{(j)}X$ represents the "j. property" of "i. instance", in the dataset. θ in Eq. (4) denotes the coefficient vector. The experimental value vector is denoted by y. Following the computation of the coefficient vector θ, its predictive value y_{pred} is calculated using Eq. (5).

$$y_{pred} = \quad X_b^{'} * \quad \theta \tag{5}$$

In Eq. (5), $X_b^{'}$ represents the feature matrix of the test dataset. 70% of the data set was used for training and 30% for testing purposes in this study. This split was made randomly. To determine the optimal coefficient vector, 10-fold cross-validation was used in the study.

2.2.2. Decision tree

The decision tree algorithm is a structure consisting of nodes and branches. The branch has a random number of nodes. Each branching node is called an internal node. Each internal node can be divided into two or more branches according to its function. The last nodes at the very end are called leaves. The values of the input variable(s) vary depending on the data set given to the tree during the training phase.

The decision tree's stimulus is an algorithm that generates a decision tree from supplied samples. By reducing the fitness function, the developed algorithm attempts to discover the best decision tree. By reducing the fitness function, the developed algorithm attempts to discover the best decision tree. Each independent variable's dataset is segmented at many points. The performed algorithm calculates the error between the predicted and actual values regarding the established fitness function at each dividing point. The split point errors are compared between the variables. The dividing point is chosen as the variable with the lowest fitness function value. Iteratively, this procedure is repeated.

Considering the data set in this study, it is seen that 11 variable parameters affect pyrolysis yield and H-content of liquid product (% wt.). For this reason, the decision tree created is quite large; it consists of many branches and leaves. The study's decision tree model is a theoretically specified regression tree, as demonstrated in Eq. (6).

$$\underset{j,s}{min}\left[\underset{c_1}{min}\sum_{x_i \epsilon S_1}(y_i - c_1)^2 + \underset{c_2}{min}\sum_{x_i \epsilon S_2}(y_i - c_2)^2\right] \tag{6}$$

The criteria feature j separates the feature space into two subspaces. (i.e. node, S_1, and S_2). x_i represents the samples. y_i is the actual value for the sample. c_i; represents the estimated value. The corresponding value is represented by i. The trial and error test was utilized in this study to discover the best criterion attribute and its corresponding value. Following the selection of the criterion feature and its corresponding value, the optimum predictive value was determined using Eqs. (7)–(8).

$$\dot{c}_1 = \frac{1}{N_1}\sum_{i=1}^{N_1} y_i(y_i \epsilon S_1) \tag{7}$$

$$\dot{c}_2 = \frac{1}{N_2}\sum_{i=1}^{N_2} y_i(y_i \epsilon S_1) \tag{8}$$

Where; N_1 and N_2 represent the number of samples from S_1 and S_2, respectively. $\dot{c}_1$ and $\dot{c}_2$ are optimal prediction labels. By repeating the procedure, a specific decision tree was produced.

Within the scope of this study, a five-depth tree structure was adopted. in both RF models created to predict the pyrolysis yield value. A five-depth tree structure in the proximate RF model, and a depth of 8 in the ultimate RF model, which was used to estimate the H-content of liquid product (%wt.) value, was used.

2.2.3. Gaussian model

A non-parametric probability model is the Gaussian regression model (GM). In comparison to deterministic models, the GM model not only forecasts the expected value on a future sample but also informs us about the model uncertainty. Consider the training set $\{(x_i,y_i);i = 1,2,.,n\}$, where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$, drawn from an unknown distribution. A GM model estimates the value of the response variable y_{new} over the model trained for the new input vector x_{new}. The mathematical expression of linear regression can be written as in Eq. (9).

$$y = x^T\beta + \varepsilon \tag{9}$$

where $\varepsilon \sim N(0,\sigma^2)$. σ^2 and β are respectively the error variance and coefficients. These values were calculated based on the data. A GM model explains the response by introducing latent variables, $f(x_i)$, $i = 1,2,.,n$, from a Gaussian model, and explicit basis functions, h. The covariance function of the latent variables captures the smoothness of the response and basis functions project the inputs x into a p-dimensional feature space.

A GM is a collection of random variables with a joint Gaussian distribution for any finite number of them. If $\{f(x),x \in \mathbb{R}^d\}$ is a GM, then given n observations $x_1,x_2,.,x_n$, the joint distribution of the random variables $f(x_1),f(x_2),..f(x_n)$ is Gaussian. The mean function $m(x)$ and covariance function $k(x,x')$ of a GM are defined. That is, if $\{f(x),x\in\mathbb{R}^d\}$ is a Gaussian model, then $E(f(x))=m(x)$ and $Cov[f(x),f(x')]= E[\{f(x)-m(x)\}\{f(x')-m(x')\}]= k(x,x')$.

$$h(x)^T\beta+f(x) \tag{10}$$

where; $f(x)\sim GM(0,k(x,x'))$. f(x) is a zero-mean GM with a covariance function. $k(x,x')$. h(x) is a set of basis functions that transform the original feature vector x in R^d into a new feature vector h(x) is R^p. β is a p-by-1 basis function coefficients vector. This model is a GM model. Eq. (11) can be used to model a response y instance. The GM model is nonparametric because a latent variable $f(x_i)$ is introduced for each observation x_i. GM model is equivalent to Eq. (12) in vector form. Eq. (14) represents the joint distribution of latent variables $f(x_1), f(x_2),., f(x_n)$ in the GM close to a linear regression model, where $K(X,X)$ looks as follows (Eq. 15). Typically, the covariance function k(x,x′) is parameterized by a set of kernel parameters or hyperparameters θ. Often $k(x,x')$ is written as $k(x,x'\downarrow\theta)$ to explicitly indicate the dependence on θ.

$$P(y_i|f(x_i),x_i) \sim N\left(y_i|h(x_i)^T\beta,f(x_i),\sigma^2\right) \tag{11}$$

$$P(y|f,X) \sim N\left(y|H\beta,f,\sigma^2 I\right) \tag{12}$$

$$X=\begin{pmatrix} x_1^T\\ x_2^T\\ *\\ *\\ *\\ x_n^T\end{pmatrix},\quad y=\begin{pmatrix} y_1\\ y_2\\ *\\ *\\ *\\ y_n\end{pmatrix},\quad H=\begin{pmatrix} h(x_1^T)\\ h(x_2^T)\\ *\\ *\\ *\\ h(x_n^T)\end{pmatrix},\quad f=\begin{pmatrix} f(x_1)\\ f(x_2)\\ *\\ *\\ *\\ f(x_n)\end{pmatrix} \tag{13}$$

$$P(f|X) \sim N(f|0,K(X,X)) \tag{14}$$

$$K(X,X)=\begin{pmatrix} k(x_1,x_1) & k(x_1,x_2) & k\ldots & k(x_1,x_n)\\ k(x_2,x_1) & k(x_2,x_2) & k\ldots & k(x_2,x_n)\\ k. & k. & k. & k.\\ k. & k. & k. & k.\\ k. & k. & k. & k.\\ k(x_n,x_1) & k(x_n,x_2) & k\ldots & k(x_n,x_n)\end{pmatrix} \tag{15}$$

2.2.4. Random forest model

The random forest (RF) model (for dealing with regression problems) is a method of creating a decision ensemble (forest) consisting of multiple decision trees. To better understand the RF model, the decision tree structure, namely Eqs. (6)–(8) should be understood. The RF model is made up of hundreds of decision trees, and the final result of the decision tree is produced by evaluating the decision results from all trees to obtain a comprehensive result.

Considering the scale of the datasets and the complexity of the model, the tree structure was created in such a way that only two parameters were optimized in the model. These parameters are; the number of trees and the maximum depth of the trees. To avoid computational complexity and to keep the computation time to a minimum, the number and depth of trees have been tried to be kept to a minimum.

A tree structure with 200 trees and a depth of 5 was used in both models created to estimate the pyrolysis yield value. To estimate the H-content of liquid product (%wt.), a tree structure with 400 trees and a depth of 8 was used in both models.

In this study, 70% of the data set used was chosen randomly as training data and 30% as test data. In addition, 10-fold cross-validation was performed to determine the best hyperparameter in the study.

2.2.5. Performance evaluation

The coefficient of determination (R^2) and Root mean square error (RMSE), Mean absolute error (MAE), Mean Squared Error (MSE) were used to analyze the performance of the models. For this purpose, Eqs. 16–19 were used in the calculation of the coefficients, respectively.

R^2 was a general indicator for all regression problems, and the more precise the model, the closer to 1. The RMSE value indicates the bias between the estimated value and the experimental value, and a suitable estimator requires a value close to zero. The mean squared error (MSE) or mean squared deviation (MSD) of an estimator (a process for estimating an unobserved variable) in statistics measures the average of the squares of the errors—that is, the average squared difference between the estimated and actual values. MSE is a risk function, corresponding to the expected value of the squared error loss. The fact that MSE is usually always strictly positive (rather than zero) is due to randomness or the estimator failing to account for information that could yield a more accurate estimate. Mean absolute error (MAE) is a statistical measure of errors between two observations reflecting the same phenomena.

$$MAE=\frac{1}{N}\sum_{i=1}^{N}|y_i-\hat{y}| \tag{16}$$

$$MSE=\frac{1}{N}\sum_{i=1}^{N}(y_i-\hat{y})^2 \tag{17}$$

$$RMSE=\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i-\hat{y})^2} \tag{18}$$

$$R^2=1-\frac{\sum(y_i-\hat{y})^2}{\sum(y_i-\bar{y})^2} \tag{19}$$

N in the equation; is the number of samples. $\hat{y}$ was the predicted value of y, $\bar{y}$ was the predicted value of y.

3. Results and discussion

3.1. Description of the datasets

The features of the liquid product dataset (a) and the hydrogen content of the liquid product dataset (b) are shown in Fig. 1(a) and Fig. 1(b), respectively. Fig. 1 depicts a box plot with the following characteristics: The bottom and top of each box represent the sample's 25th and 75th percentiles, respectively. The interquartile range is the distance between the bottom and top of each box. The sample median is represented by the red line in the center of each box. The plot displays sample skewness if the median is not centered in the box. Whiskers are lines that extend above and below each box. Whiskers extend from the interquartile range's farthest observation to the whisker length's farthest observation (the adjacent value). Outliers are observations that extend beyond the whisker length. An outlier is defined by default as a value that is more than 1.5 times the interquartile range away from the bottom or top of the box. The outliers are represented by a blue o sign. Comparing box plot medians is analogous to performing a visual hypothesis test, such as the t-test for means.

The ash, fixed carbon, and volatile matter contents presented are also given on a moisture-free (dry) basis since the comparison of biomass properties is much more meaningful and accurate this way.

For the liquid product yield (%wt.) dataset and the hydrogen content of the liquid product dataset, it was observed that the proximate content of biomass samples i.e., ash, fixed carbon, and volatile matter contents changed in the ranges of 0.35%– 17.33; 9.59%– 31.13%; 51.45%– 87.34% and 0.1%– 15.43%;7.3%– 18.0%;62.43%– 86.0%, respectively. For the liquid product yield (%wt.) dataset and the hydrogen content of liquid product dataset, the ultimate content i.e., C–H–N-O were in the range of 39.34% – 60.46%; 44.82 – 62.10%, 5.08% – 9.08%; 5.08 – 10.15%, 0.22% – 9.29%; 0.30 – 9.29% and 27.36% – 53.3%; 24.90 – 549.18%, respectively. As a result, it is understood

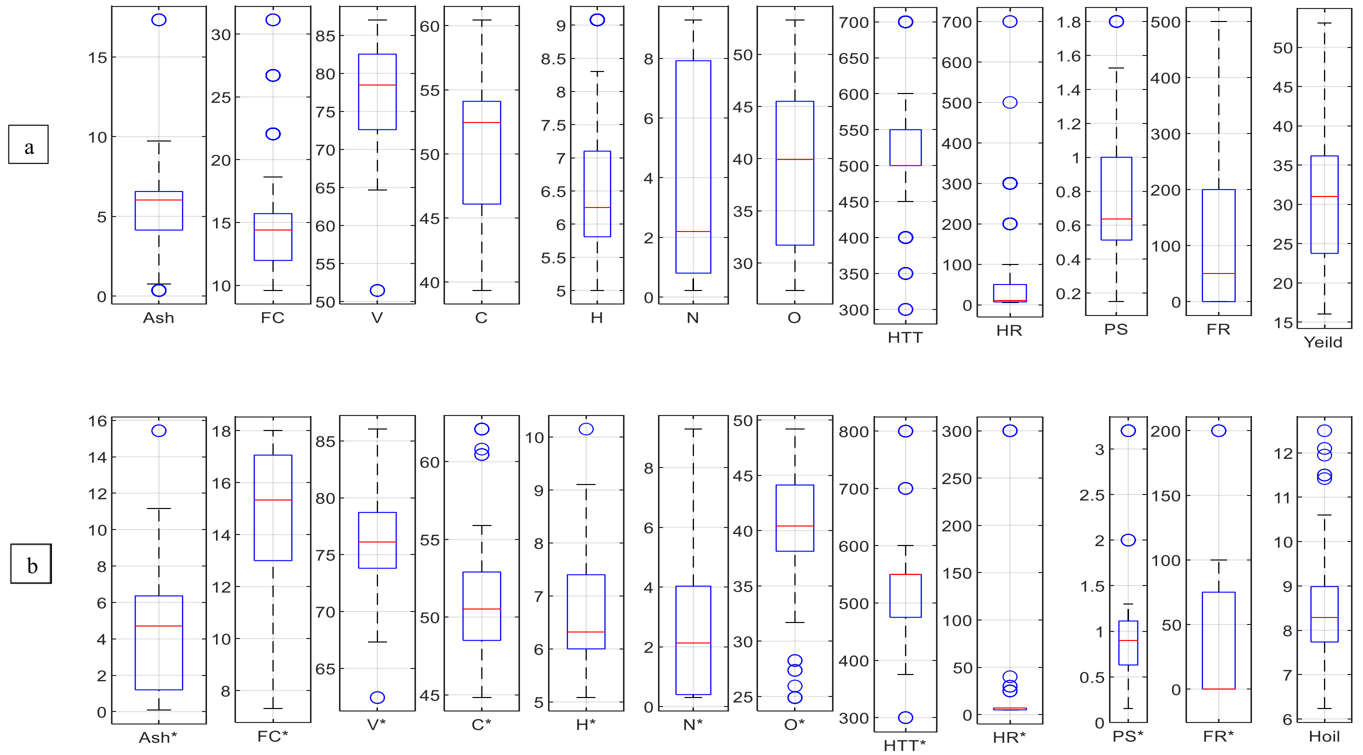


Fig. 1. Details on the liquid product yield dataset (a) and the hydrogen content of the liquid product dataset (b).

that the dataset samples were selected to include all biomass species known to have lignocellulosic structure when the proximate-ultimate compositions of the biomass samples are examined. This is an important implication for the success of the dataset. It was concluded that the proximate and ultimate composition differences observed between the two data sets were primarily due to the presence of a large number of algae in the liquid product data set. The heating rate and pyrolysis temperature ranges were not significantly wider, owing to the fixed-bed pyrolysis reactor conditions. For the liquid product yield (%wt.) and the hydrogen content of the liquid product dataset, the pyrolysis conditions (i.e., the highest pyrolysis temperature (HTT), heating rate (HT), particle size (PS), the nitrogen flow rate (FR)) were in the range of $300-700\,^{\circ}$C; $300-800\,^{\circ}$C, $5-700\,^{\circ}$C/min; $5-300\,^{\circ}$C/min, $0.15-1.8$ mm; $0.15-3.20$ mm, and $0-500$ mL/min; $0-200$ mL/min, respectively. For the biomass diversity content presented in the data sets, in the pyrolysis conditions expressed, it was observed that the liquid product yield (%wt.) and the H-content of liquid product yield (% wt.) were a range of 16.05%– 53.07% and 1.85%– 12.5%, respectively.

3.2. Analyzing the linear relationship between two variables statistically

The Pearson correlation coefficients (PCCs) of the two variables are shown in Table 2. Since the proximate analysis' fixed carbon content (FC=100-(V +Ash)) and the ultimate analysis' oxygen content (O=100-(C+H+N)) were calculated using these differences equations. Because of this, the oxygen content and fixed carbon inputs were removed from the MLR models when they were created.

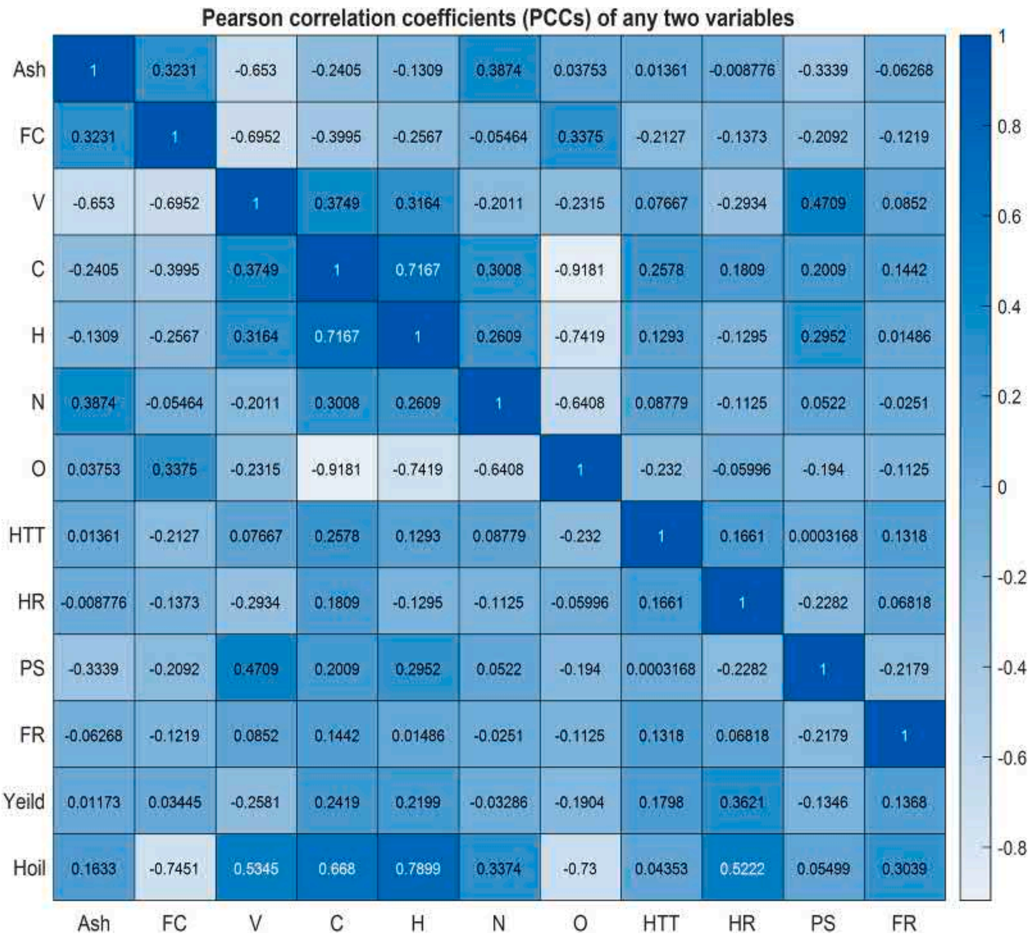
Table 2 shows that there were no obvious linear relationships between other variables and liquid product yield. The same result was validated the H-content of liquid product yield. The PCCs, on the other hand, revealed that the bulk of the variables had no evident relationships with one another. Thus, it was concluded that a more detailed MLR model and RF model should be used. Therefore, different models were used in this study. The outcome of a t-test between any two variables is given in Table 3.

3.3. The RF Models, Gaussian Models (GM), and Decisions Tree Model (DT) performance prediction for the liquid product yield (%wt.) and its hydrogen content

The optimum hyper-parameters determined for each model are shown in Table 4. Grid searching and 5-fold cross-validation of training sets were used to find the parameters. The random forest model (RF), Gaussian model (GM), and Decisions tree model (DT) are among these models. The results demonstrate that ultimate models had higher tree numbers than proximate models, which makes sense given the larger number of characteristics. The maximum depths in both the ultimate and proximate models were the same, owing to similar distinguishing features in both. The test sets have covered 20% of the experimental data used to validate the performances of all models' trained models. Fig. 2 and Fig. 3 shows a detailed comparison of experimental data and estimated data for the MLR(a), DT(b), GM(c), and RF(d) models. The experimental data are represented on the x-axis, the estimated value is represented on the y-axis, and the straight line shows the best prediction condition, where n is equal to the experimental data.

The R^2, RMSE, MSE, MAE, prediction speed, training time, and model scores are listed in Table 5. The MRL model's regression coefficients were determined to be less than one for each comparison. The regression coefficient ranged from 0.19 to 0.6. The MLR performed less well than the other models, according to these findings. This finding indicated that there is no straightforward linear link between the pyrolysis liquid product yield and H-content of liquid product (%wt.) and the variables studied. In comparison to other estimations, the MLR model performs better in Ultimate-H and Proximate-H estimation. It outperformed the competition (R^2 =0.6). It was thought that these results were more consistent, because of the limited linear relationship between biomass hydrogen concentration and PCC analysis. Because of the limited linear relationship between the hydrogen content of biomass sources and the hydrogen content of the pyrolysis liquid product, the PCC analysis produced more consistent results. In addition, The RF model has been determined as the model with the best performance. The regression coefficients of the model are found in the range of 0.82–0.94.

Table 2
Coefficients of Pearson correlation (PCC) of any two variables.



Pearson correlation coefficients (PCCs) of any two variables

	Ash	FC	V	C	H	N	O	HTT	HR	PS	FR
Ash	1	0.3231	-0.653	-0.2405	-0.1309	0.3874	0.03753	0.01361	-0.008776	-0.3339	-0.06268
FC	0.3231	1	-0.6952	-0.3995	-0.2567	-0.05464	0.3375	-0.2127	-0.1373	-0.2092	-0.1219
V	-0.653	-0.6952	1	0.3749	0.3164	-0.2011	-0.2315	0.07667	-0.2934	0.4709	0.0852
C	-0.2405	-0.3995	0.3749	1	0.7167	0.3008	-0.9181	0.2578	0.1809	0.2009	0.1442
H	-0.1309	-0.2567	0.3164	0.7167	1	0.2609	-0.7419	0.1293	-0.1295	0.2952	0.01486
N	0.3874	-0.05464	-0.2011	0.3008	0.2609	1	-0.6408	0.08779	-0.1125	0.0522	-0.0251
O	0.03753	0.3375	-0.2315	-0.9181	-0.7419	-0.6408	1	-0.232	-0.05996	-0.194	-0.1125
HTT	0.01361	-0.2127	0.07667	0.2578	0.1293	0.08779	-0.232	1	0.1661	0.0003168	0.1318
HR	-0.008776	-0.1373	-0.2934	0.1809	-0.1295	-0.1125	-0.05996	0.1661	1	-0.2282	0.06818
PS	-0.3339	-0.2092	0.4709	0.2009	0.2952	0.0522	-0.194	0.0003168	-0.2282	1	-0.2179
FR	-0.06268	-0.1219	0.0852	0.1442	0.01486	-0.0251	-0.1125	0.1318	0.06818	-0.2179	1
Yeild	0.01173	0.03445	-0.2581	0.2419	0.2199	-0.03286	-0.1904	0.1798	0.3621	-0.1346	0.1368
Hoil	0.1633	-0.7451	0.5345	0.668	0.7899	0.3374	-0.73	0.04353	0.5222	0.05499	0.3039

Table 3
The outcome of a t-test between any two variables.

	Ash	FC	V	C	H	N	O	HTT	HR	PS	FR
Ash	1	0	0	0	0	0	0	0	0	0	0
FC	0.20	1	0	0	0	0	0	0	0	0	0
V	0.72	0.76	1	0	0	0	0	0	0	0	0
C	0.36	0.50	0.26	1	0	0	0	0	0	0	0
H	0.25	0.37	0.20	0.65	1	0	0	0	0	0	0
N	0.27	0.18	0.32	0.18	0.14	1	0	0	0	0	0
O	0	0.22	0.35	0.94	0.79	0.71	1	0	0	0	0
HTT	0.11	0.33	0	0.14	0	0	0.35	1	0	0	0
HR	0.14	0.26	0.41	0	0.25	0.24	0.19	0	1	0	0
PS	0.44	0.33	0.37	0	0.17	0	0.31	0.13	0.35	1	0
FR	0.19	0.25	0	0	0.11	0.15	0.24	0	0	0.34	1
Liquid P yield (%wt)	0.12	0	0.37	0.12	0	0.16	0.31	0	0.25	0.26	0
H-oil yield (%wt)	0.14	0.16	0.14	0.36	0.34	0	0	0.30	0.47	0	0.26

Table 4
Grid-Search Method Optimum Hyper-Parameters in Gaussian Models (GM), RF Models, and Decisions Tree Model (DT).

	GM Models					RF Models		DT Models
	Basıc Function	Kernel Function	Kernel Scala	Signal Standard Deviation	Sigma	Trees Number	Max Depth	Max Depth
Proximate-Yield	Constant	Quadratic	50.541	5.738	5.738	200	5	5
Proximate-H-oil	Constant	Quadratic	36.134	1.0005	1.0005	400	8	8
Ultimate-Yield	Constant	Quadratic	44.684	5.738	5.738	200	5	5
Ultimate-H-oil	Constant	Quadratic	32.047	1.0005	1.0005	400	8	5

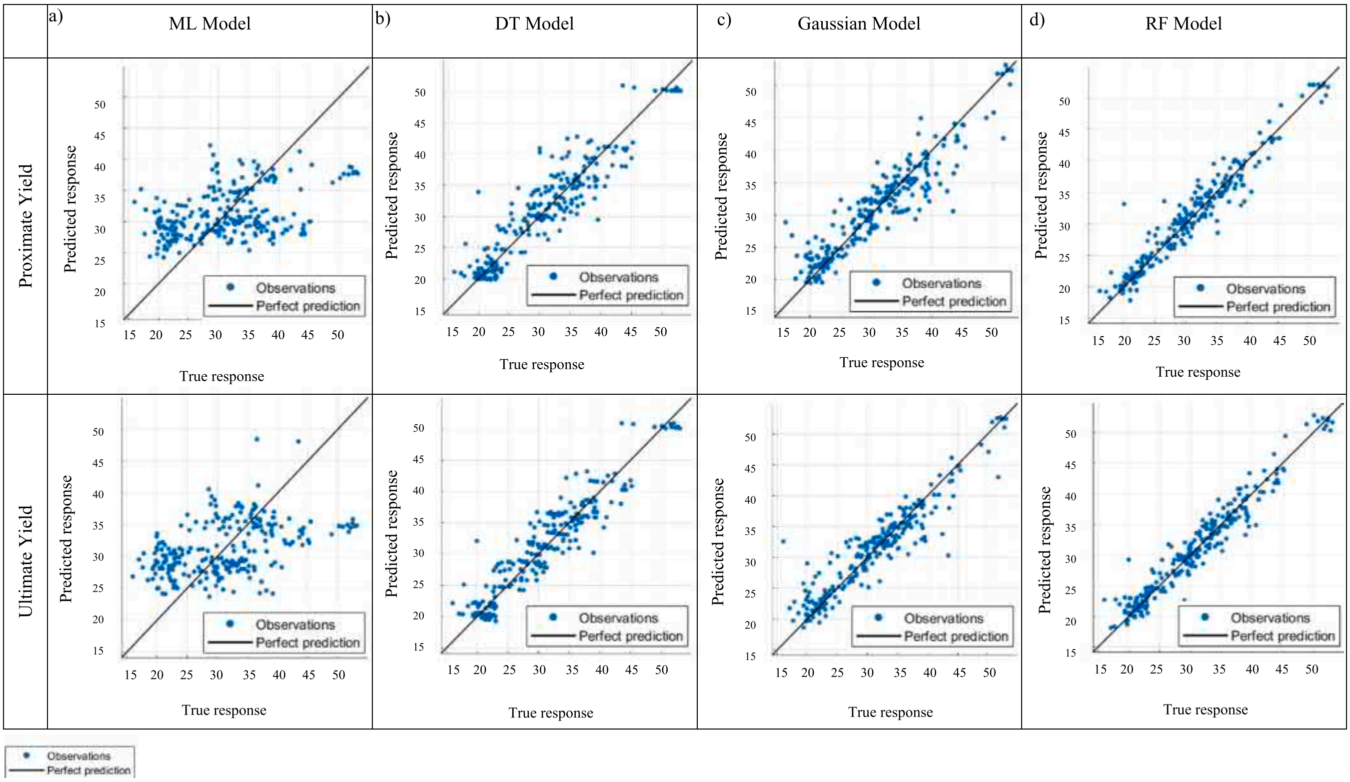


Fig. 2. Prediction and actual response graphs for liquid product yield a) MLR b) DT c) GM d) RF models.

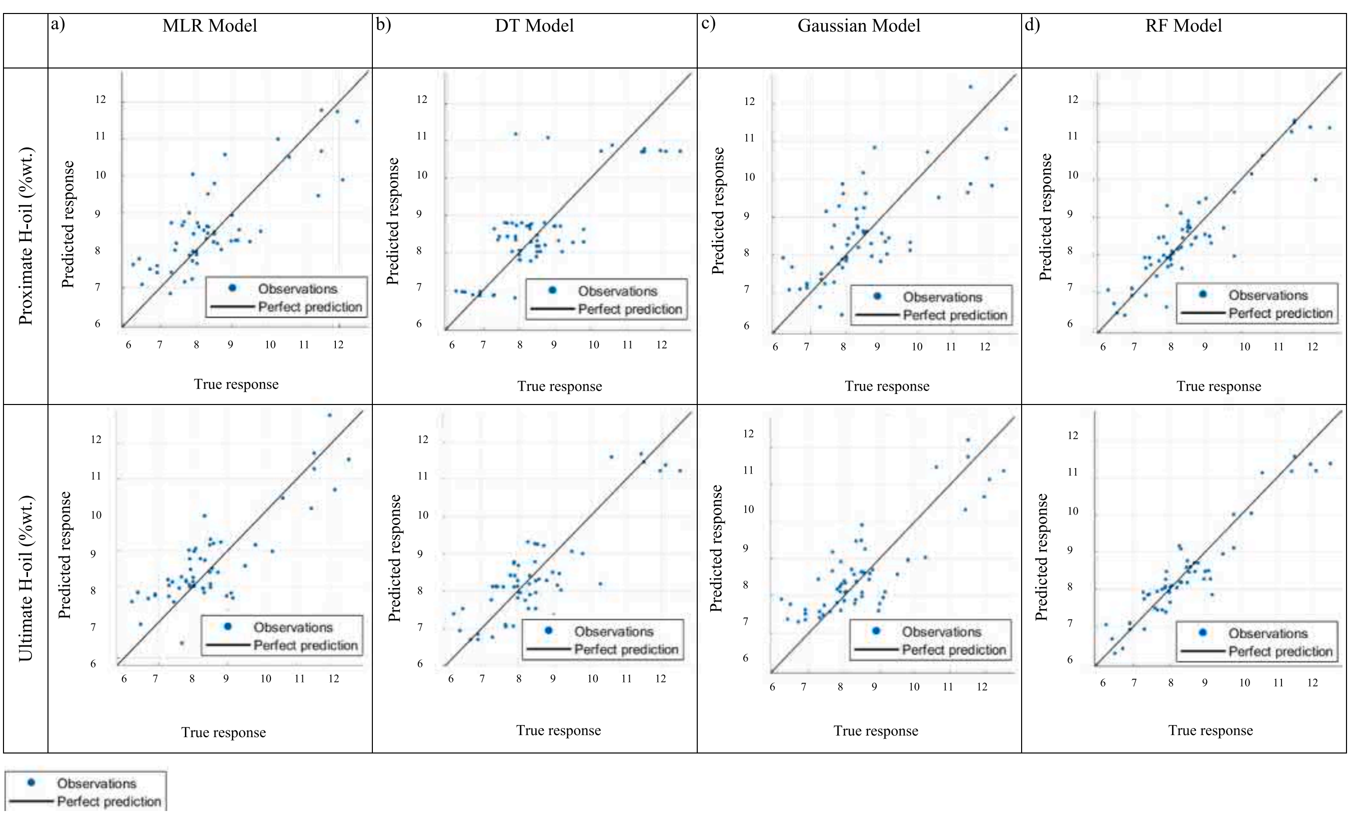


Fig. 3. Prediction and actual response graphs for H-content of liquid product yield a) MLR b) DT c) GM d) RF models.

Table 5
R^2, RMSE, MSE, MAE, Prediction Speed, Training Time, Scores of Models.

Model		R^2	RMSE	MSE	MAE	Prediction Speed (obs/sec)	Training Time (sec)
MLR Models	Proximate-Yield	0.19	7.3132	53.483	5.9090	5300	0.5871
	Proximate- H-oil	0.6	0.9066	0.8219	0.7362	910	0.4857
	Ultimate-Yield	0.2	7.2577	52.674	5.9897	5100	0.5438
	Ultimate –H-oil	0.6	0.9086	0.8256	0.7392	1400	0.4883
DT Models	Proximate- Yield	0.83	3.3658	11.329	2.4460	12,000	0.4003
	Proximate- H-oil	0.65	0.8521	0.7261	0.6211	3400	0.3400
	Ultimate-Yield	0.88	2.8056	7.8715	2.1391	12,000	0.3712
	Ultimate -H-oil	0.67	0.8274	0.6847	0.6317	3800	0.3361
GM Models	Proximate- Yield	0.86	3.0648	9.3927	2.1426	10,000	1.5877
	Proximate- H-oil	0.56	0.9540	0.9102	0.7247	3100	0.6919
	Ultimate-Yield	0.88	2.8755	8.2684	1.9173	9800	2.3105
	Ultimate -H-oil	0.68	0.8146	0.6636	0.6540	3300	0.7211
RF Models	Proximate- Yield	0.93	2.0762	4.3104	1.5481	650	9.7046
	Proximate- H-oil	0.82	0.6108	0.3722	0.4311	95	17.194
	Ultimate-Yield	0.94	2.0205	4.0823	1.5135	690	8.9752
	Ultimate -H-oil	0.90	0.4577	0.2068	0.3480	89	17.436

The predictive success of RF models was higher, as a result of comparative large-scale correlated data sets. As a whole, the success of all RF models is > 0.82. Both the regression coefficients and the RMSE values, which are relatively close to zero, prove that an acceptable performance has been achieved with the RF model. The success rate was discovered to be greater than that of the ANN and support vector machine models [18]. As a result, it was determined that RF was needed to determine the complex relationship between factors and targets.

In addition, the results showed that the success of predicting pyrolysis liquid product yield and liquid product H-content was higher based on the ultimate analysis data for all models. The success obtained with the proximate analysis data remained below the ultimate analysis data.

When the regression coefficients presented in Table 5 are examined, it is observed that the estimation made with the final analysis data can be used with higher accuracy in estimating the H-content of the liquid product. The regression coefficients for the RF, GM, and DT models were determined as 0.9, 0.68, and 0.67, respectively.

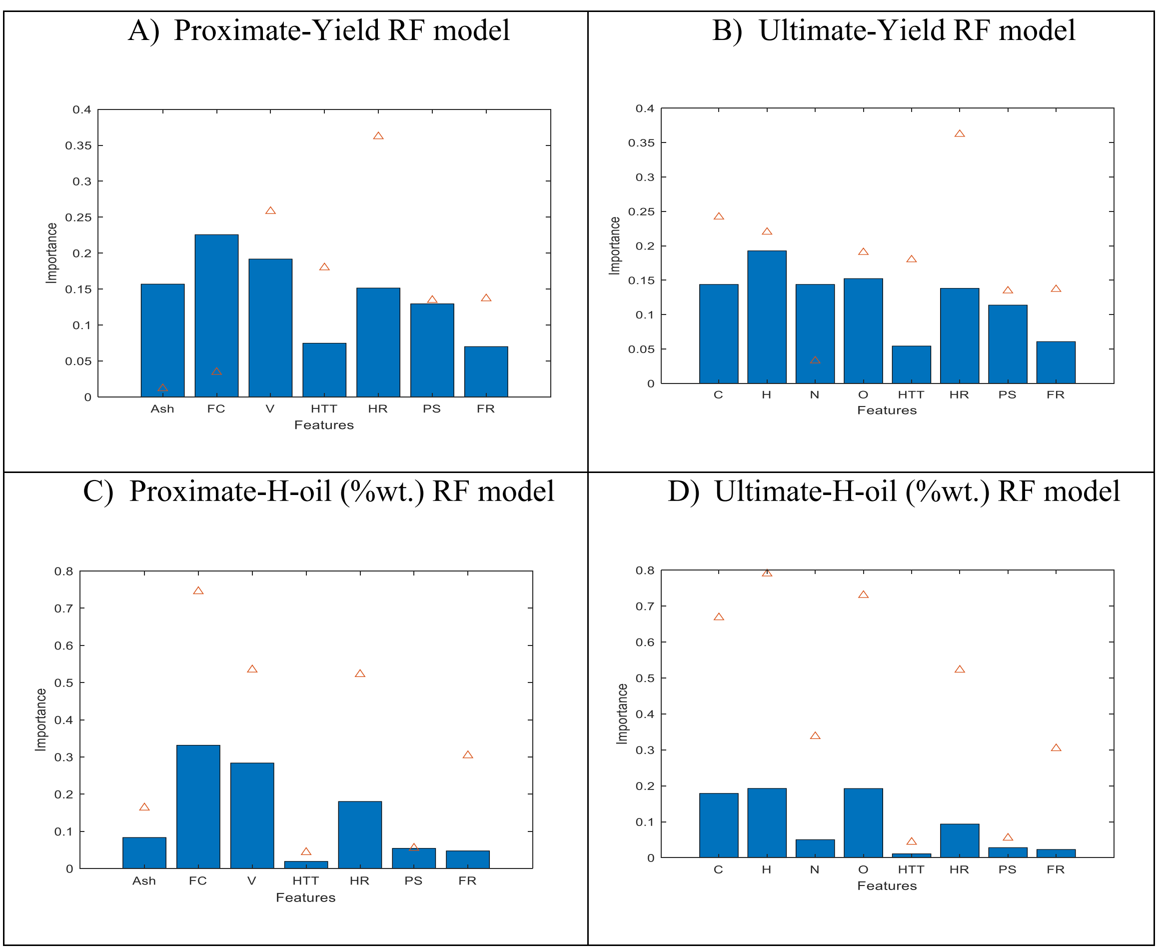


Fig. 4. PCCs comparison and Feature importance of a) Proximate-yield RF model b) Ultimate-yield RF model c) Proximate-H-oil (%wt.) RF model b) Ultimate-H-oil (%wt.) RF model.

3.4. Feature importance

Importance feature information was obtained by using the average pollution reduction method in the whole models of RF. The Importance feature values of each model are presented in bar graph form and the corresponding value of PCC is presented with the triangle symbol in Fig. 4. To provide a suitable comparison, the absolute values of the PCC values were used.

It should be noted that the importance of the feature was represented a nonlinear relationship, while the PCC values were represented the linear relationship. In general, it was observed that the results of Pearson analysis and RF analysis contained large inconsistencies for other variable parameters. This result proved that there were no simple linear relationships between the liquid product yield and H-content of liquid product (%wt) and the variables. Besides, as a result of PCC and RF analyses of the proximate-yield model and ultimate-yield model, it was determined that the HR change was the most critical pyrolysis condition parameter affecting the liquid product yield (%wt) and The H-content of liquid product yield (%wt.). Besides the PCC and RF analyses of the ultimate-yield model, it was determined that the H content of biomass resources was the most critical parameter affecting the liquid product yield (%wt) and the H-content of liquid product (%wt). In addition, the effect of the proximate FC-content of biomass sources on the liquid product yield (%wt) was higher than the other parameters. As a result of the estimation made with the RF model using the proximate analysis data, the composition has an effect of approximately 60% and the pyrolysis conditions of 40% on the liquid product yield (%wt.). The H-content of liquid product yield (%wt.) has been affected by these parameters by approximately 70% and 30%, respectively. As a result of the estimation made with the RF model using the Ultimate analysis data, it was determined that the composition has an effect of approximately 65% and the pyrolysis conditions by 35% on the liquid product yield (% wt.). The H-content of liquid product (%wt.) has been affected by approximately 63% and 37%, respectively. As can be seen from these results, the biomass composition affects the liquid product yield and its hydrogen content more than the pyrolysis conditions. Compared with the pyrolysis final temperature, it has been determined that the rate of reaching the final temperature, that is, the heating rate, affects the pyrolysis liquid product yield more. This result is in agreement with the literature. In biomass pyrolysis, the heating rate is an important parameter that affects the distribution and chemical composition of pyrolysis products. However, the heating rate should be considered as a function of the residence time and temperature. Accordingly, high liquid product yield can be obtained at low heating rates and low temperatures, short and medium retention times; it is known that liquid and solid product yields are close to each other, and gaseous product yield is low in long retention times under the same conditions [19]. In addition, it is observed that the effect of PS (>%12) and FR (>%6) on the pyrolysis liquid product yield is more than its contribution to the hydrogen content of the product.

3.5. Partial dependence analysis for liquid product yield (%wt.) and H-content of liquid product (%wt.): one-way and two-way partial dependence

A detailed partial dependency analysis was performed to investigate in-depth the relationship between biomass composition and pyrolysis conditions and target variables (pyrolysis liquid product yield and product hydrogen content). The effect of the parameters, the effects of which were examined, on the pyrolysis liquid product and the hydrogen content of the liquid product, the way and amount of influence, partial dependence analysis were tried to be revealed. The main functioning of the partial dependency analysis is to limit the parameters other than the investigated parameter to mean values and to estimate the effect of the only parameter whose effect is examined on the target [17]. (ie, the liquid product yield (%wt.) or the hydrogen content of the liquid product).

In Fig. 5(a-k) and Fig. 6(a-k), the effect of biomass composition and pyrolysis conditions on liquid product yield (%wt.) and H-content of liquid product (%wt) as a result of partial dependency analysis is presented. Fig. 7(a-n) and Fig. 8(a-n) show the two-way partial dependence plots for liquid product yield (%wt.) and H-content of liquid product (% wt.), respectively. It is known that the ash content of biomass sources (in the range of 3–6%) acts as a catalyst due to alkali and alkaline earth metal contents and increases the yield of liquid products. In addition, it is known that silica and alkali metals in the structure of the ash prevent the circulation of the mobile phase in the furnace. When Fig. 5(a) is examined, the result obtained is by the literature information [1,20]. In Table 1, coefficients of Pearson correlation (PCC) of ash and liquid product yield (%wt.) were found to be positive, that is, it was found that there was a right relationship between the ash content and the liquid product yield (%wt.). These results concluded PCC analysis was correct to evaluate.

When Fig. 5(b) is examined, it is determined that the liquid product yield (%wt.) of biomass sources containing higher than 12–16% FC is lower. This figure, FC, and liquid product yield (%wt.) indicate a negative correlation for this range of values. But, the Pearson correlation coefficient (PCC) of FC and liquid product yield (%wt.) was found to be positive in Table 1. This result is not consistent with Fig. 5(b). These results concluded PCC analysis was not correct to evaluate. Also, it is seen in Fig. 5(c) that the yield of the liquid product increases as the volatile matter ratio of biomass decreases. In other words, VC and liquid product yield (%wt.) indicate a negative correlation. This result is not consistent with the studies in the literature [21,22]. It can be thought that this inconsistency is the result of a lack of sampling, or it can be said secondary reactions and re-polymerization reactions lead to the reduced liquid product yield (%wt.). Besides, it is an expected and acceptable result that as the volatile matter content of biomass sources increases, the fixed carbon content decreases. It is an expected result that as the heating rate in the pyrolysis process of biomass sources with high volatile substance content increases, the volatile substance content supports the formation of tar, thus increasing the liquid product efficiency [23]. A similar situation applies to the ash content ie the organic matter content decreases as the ash content increases. And the high heating rate and ash content are made a positive impact on the liquid product efficiency due to the increase in catalytic reactions [24]. According to the grasp, we thought that the hydrogen content of biomass is the key to liquid product quality. In addition, one-way partial dependency plots between H-content of liquid product (%wt.) and biomass composition are presented in Fig. 6(a)-(g). When evaluated within the biomass composition parameters, it was seen that the H-content of biomass had the most effect on the H-content of liquid product (%wt.). High hydrogen-containing biomass sources have high H-content liquid product (%wt.)s. There is a positive correlation between the two parameters. This result is in agreement with both the literature and the PCC analysis results (PPC; 0.7899). A group researcher, who examined Euphorbia rigida, sunflower pulp, and hazelnut shells, stated that these plants are rich in aromatic structures; reported that aromatic ring compounds (methylene and naphthenic proton) increased the ultimate hydrogen content, and as a result, higher H-content of liquid product (% wt.) was achieved as a result of pyrolysis processing. In the study, it was emphasized that a higher quality liquid fuel was obtained [25]. Due to the positive correlation between nitrogen and protein, of which the most by hydrogen, biomass sources have high nitrogen contents. This situation is understandable. Also, the positive PPC value (0.3374) is in agreement with the increase in H-content of liquid product (%wt) in Fig. 6(f).

Fig. 3 shows that the heating ratio (HR) is the most influential pyrolysis condition (Fig. 4), and Table 1 supports Fig. 4. Table 1 shows positive correlations between the HR and liquid product yield (%wt.) and its hydrogen content. According to Ertaş and Alma (2010)'s [26] investigation report, at higher HR values, the time to reach the final

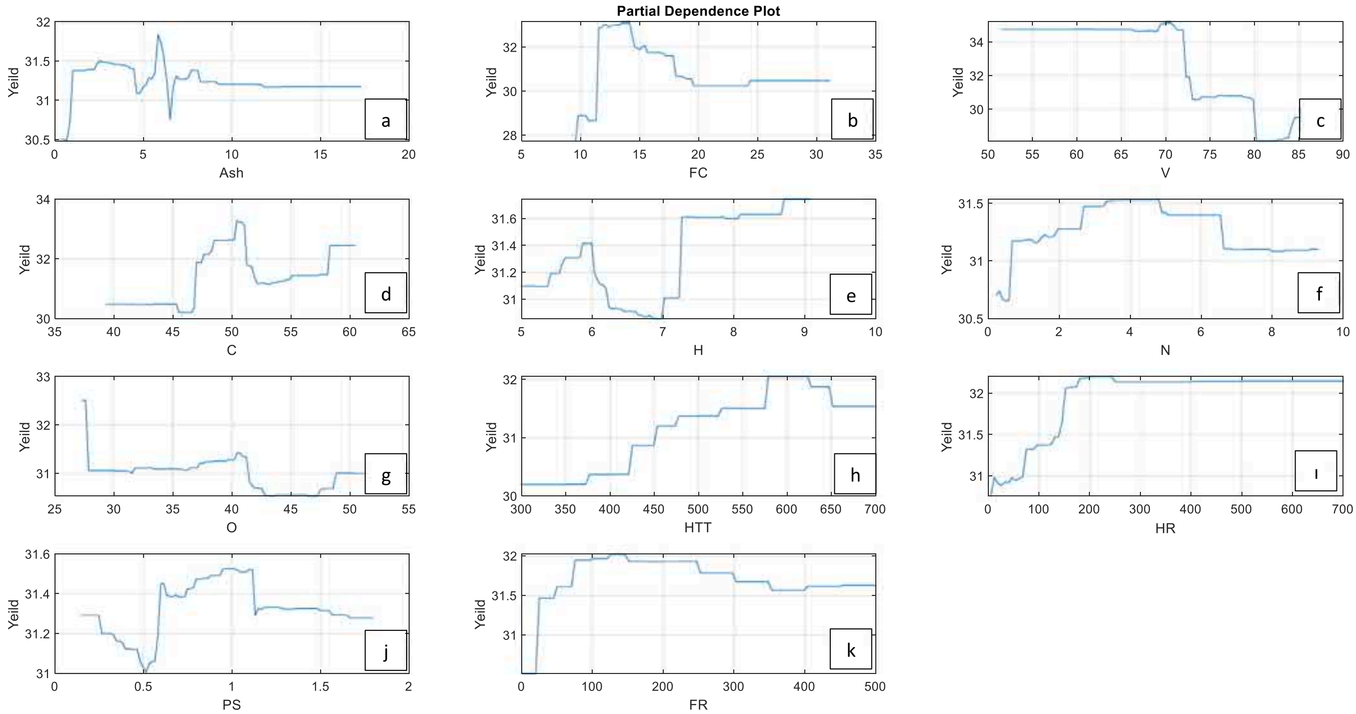


Fig. 5. Model of liquid product yield (%wt.) with one-way partial dependence plot.

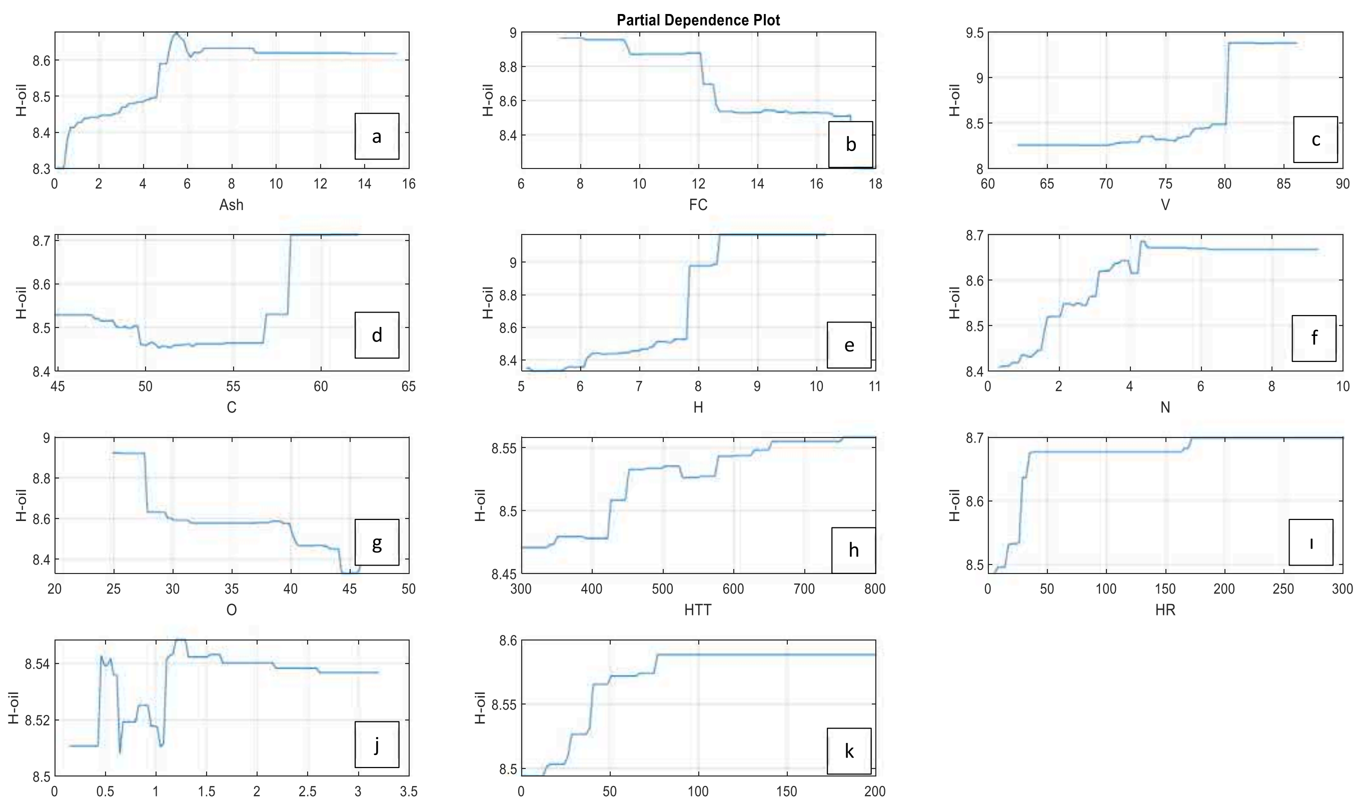


Fig. 6. Model of H-content of liquid product (%wt.) with one-way partial dependence plot.

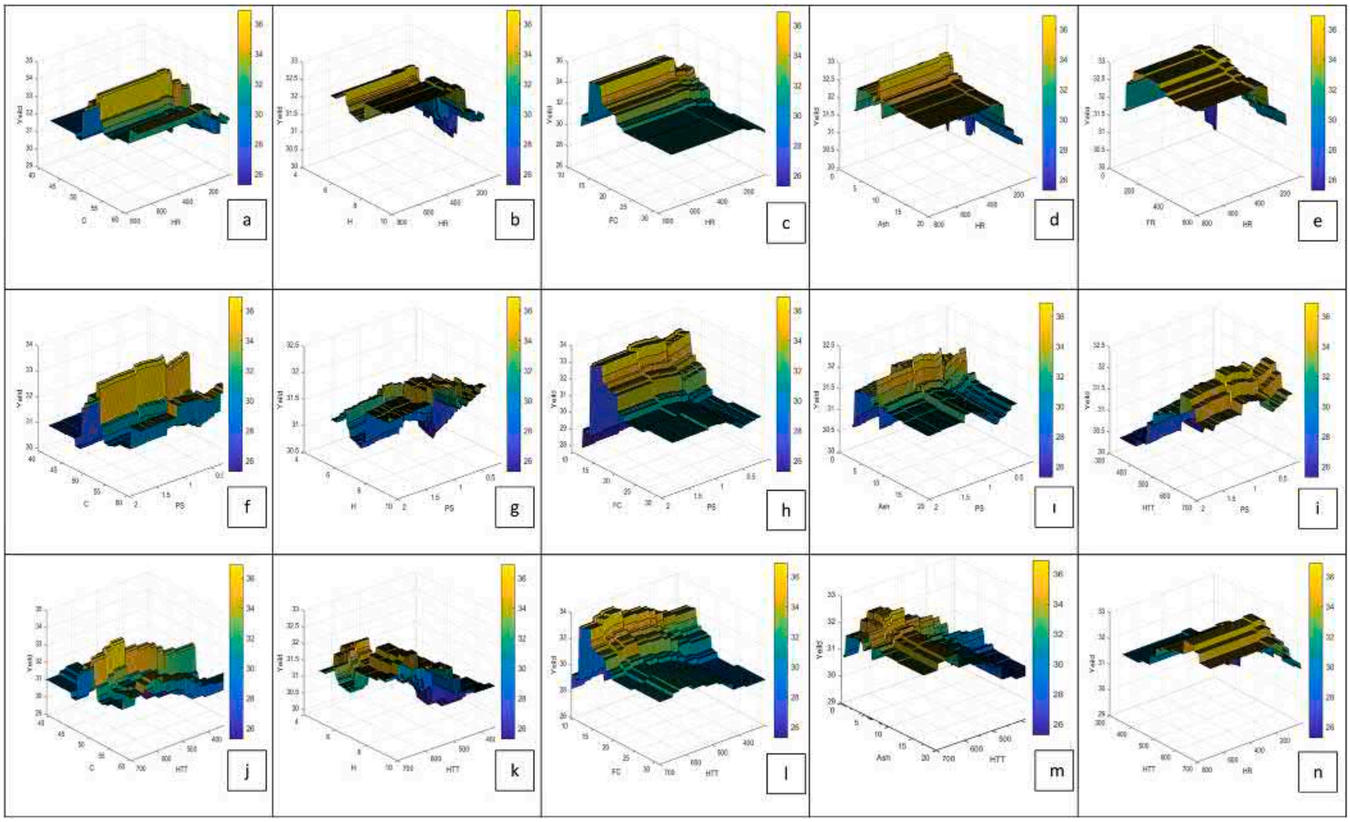


Fig. 7. Model of liquid product yield (%wt.) with two-way partial dependence.

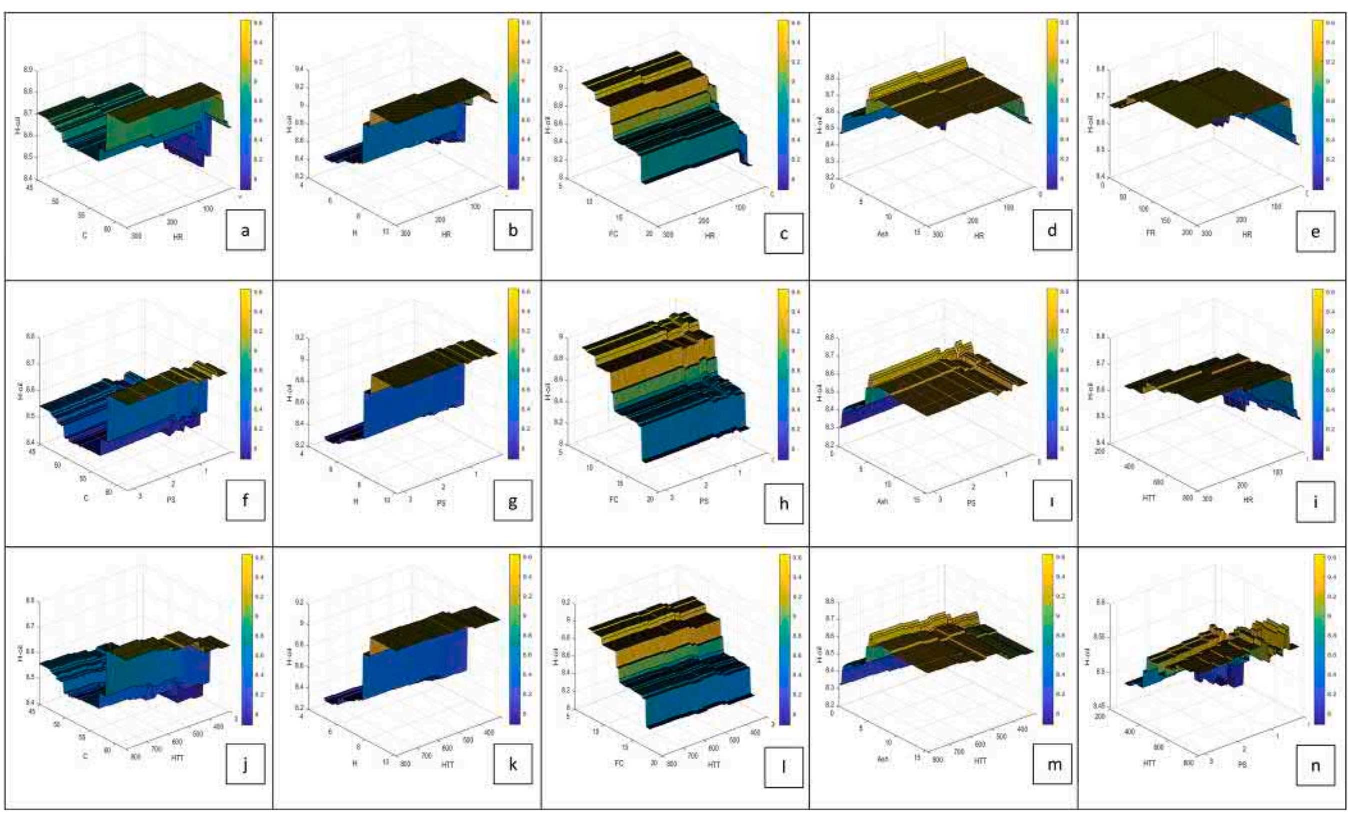


Fig. 8. Model of H-content of liquid product (%wt.) with two-way partial dependence plot.

temperature is shorter, ie the residence time is shorter. It is known that the gas product yield increases with the increase in residence time, and the primary pyrolysis products turn into lower molecular gas products by thermal decomposition [27]. Therefore, the liquid product yield is obtained higher with high heating rates. Fig. 5(ı) and 6(ı) show that the maximum liquid product yield and its h-content were obtained at the heating value of above 200 °C/min.

Liquid product yield (%wt.) with HTT shows a positive correlation at HTT< 600 and a negative correlation at HTT> 650 (Fig. 5(h)). At high temperatures, the temperature difference between the particle and the environment is higher. It is thought that this condition increased the heat transfer rate between the particle and the environment and the temperature difference between the center and the surface decreased. Therefore, it can be said that most of the volatile components in the biomass structure are separated from the particle. This situation also causes a decrease in the yield of solid products. It is known that the liquid product yield increases as the pyrolysis temperature increases and the liquid product yield reaches its maximum value at 500–600 C. But, the yield of the liquid products decreases above the value. Because the volatile components decompose at high ambient temperatures and turn into gaseous products with smaller molecules [28–30].

There is a complex correlation between PS and liquid product yield (%wt.) (Fig. 5(i)) and hydrogen content of the liquid product (Fig. 6(i)). Among the expressed variables, there are positive correlations in the 0.5 − 1 mm range and negative correlations in PS> 1 mm. This situation is the result of the mass transfer resistance between phases.

As is shown in Fig. 5(k) and Fig. 6(k), higher FR values (0 <FR<100) enhanced liquid product yield (%wt.) and its hydrogen content. The observed increase in the pyrolysis liquid product yield in the presence of the entraining inert gas is a result of the faster removal of volatile components by the entraining inert gas from the pyrolysis system. Thus, the occurrence of secondary reactions such as thermal cracking, partial oxidation, depolymerization, and recondensation is prevented [31,32].

In addition to the comments given above, two-way partial dependence plots (Fig. 7 and Fig. 8) show some synergic conclusions. Agglomeration would be a problem due to the higher pyrolysis temperatures and small particle size. However, at low final temperatures, apparent activation energies could not be afforded. In addition to the points made above, it is common knowledge that greater pyrolysis temperatures and small particle sizes result in agglomeration. However, at low final temperatures, the need for apparent activation energies cannot afford. A moderate HTT combined with a higher heating rate has a good synergetic effect on oil yield. The effects of PS and HTT on the H-content of liquid products are fairly limited (%wt.). Between HTT and PS, there is a limited synergetic effect. For lower temperatures (300–500 °C), the moderate particle size (1.0–1.5 mm) may contribute to the H-content of the liquid product (percent wt.). However, for the higher temperature values (>600 °C), a small particle size (0.5–1.0 mm) is required.

4. Conclusions

In this study, the relationship between liquid product yield (%wt.) and hydrogen content of liquid product with the composition of biomass and pyrolysis conditions was tried to be revealed. For this purpose, linear and nonlinear relationships between variable parameters and liquid product yield (%wt.) and hydrogen content were investigated. The result of the work carried out; it was determined that the Pearson correlation coefficient (PCC) analysis and the multiple linear regression were not sufficient to analyze the relationships. It was concluded that liquid product yield (%wt.) and hydrogen content of the liquid product can be predicted with a high success rate with the random forest (RF) model. In addition, partial dependency analysis was performed in the study, so that nonlinear relationships were specifically revealed and interpreted. Although partial dependency analysis is widely used in engineering applications in general, it is a new method to investigate product yield and quality depending on biomass composition and pyrolysis conditions. There is a very limited number of studies such as [17]. Although the method using the previous study gives successful results, studies and efforts should continue to reveal the relationship between the variables in more detail. In this context, in this study, the amount of liquid product and the hydrogen content (quality) of the liquid product was optimized depending on the biomass composition and pyrolysis conditions. The success rates of Proximate-Yield, Ultimate-Yield, Proximate-H, and Ultimate-H models, which Thang et al. [17] suggested using RF classifier in their study, were 92%, 87%, 79%, and 84%, respectively. Considering Table 4, 93%, 94%, 82%, and 90% success rates were achieved with the models we suggested. With the models proposed in this study, according to the R2 performance criteria, a success increase of 1% in the Proximate-Yield model, 7% in the Ultimate-Yield model, 3% in the Proximate-H model, and 6% in the Ultimate-H model was achieved [17].

As a result; it was determined that the composition of the biomass (proximate and ultimate composition) had a greater effect on both the yield and hydrogen content of the liquid product obtained as a result of the pyrolysis process than the pyrolysis conditions. To increase liquid product yield (%wt.) and hydrogen content, it has been determined that biomass sources with high hydrogen content should be preferred and that pyrolysis operation should be made at high heating rates (>150 °C/min) at approximately 550 − 600 °C. It was also observed that while the nitrogen flow rate increased, the liquid product yield (%wt.) and hydrogen content was increased. But; it was concluded that these rates of increase are very low and have a limited effect.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jaap.2022.105546.

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